



An Overview of Statistical Learning

Statistical learning refers to a vast set of tools for *understanding data*. These tools can be classified as *supervised* or *unsupervised*. Broadly speaking, supervised statistical learning involves building a statistical model for predicting, or estimating, an *output* based on one or more *inputs*. Problems of this nature occur in fields as diverse as business, medicine, astrophysics, and public policy. With unsupervised statistical learning, there are inputs but no supervising output; nevertheless we can learn relationships and structure from such data. To provide an illustration of some applications of statistical learning, we briefly discuss three real-world data sets that are considered in this book.

Wage Data

In this application (which we refer to as the *Wage* data set throughout this book), we examine a number of factors that relate to wages for a group of men from the Atlantic region of the United States. In particular, we wish to understand the association between an employee's *age* and *education*, as well as the calendar *year*, on his *wage*. Consider, for example, the left-hand panel of Figure 1.1, which displays *wage* versus *age* for each of the individuals in the data set. There is evidence that *wage* increases with *age* but then decreases again after approximately age 60. The blue line, which provides an estimate of the average *wage* for a given *age*, makes this trend clearer. Given an employee's *age*, we can use this curve to *predict* his *wage*. However, it is also clear from Figure 1.1 that there is a significant amount of variability associated with this average value, and so *age* alone is unlikely to provide an accurate prediction of a particular man's *wage*.